

## Standing Waves and Harmonics, Part 2

We have looked at the standing waves when a string is tied down at both ends, which covers the behavior of a wide range of instruments. We'll next look at the behavior of a string with an open end. Your mental picture here should be a string attached to a rod at one end. The connection to the rod is frictionless, so the string can freely slide up and down. I'll describe how this works, but will only hold you responsible for the end result.

### 1. Reflection of a Free End

The lack of friction at the end means that the only force between the string and the rod must be perpendicular to the rod. Since there's no friction, if you try to apply a vertical force, the string just slides along the rod. In more detail, if you zoom in to where the string hits the rod, if the string hits the rod at an angle, there will be a vertical force from the tension in the string. Since the rod is frictionless, you can't balance that vertical force out. As you zoom further in, the vertical force stays constant because the angle the string makes is constant, but the length of the string goes down and so the mass goes down. The net effect since  $F = ma$  is that if  $F$  is fixed and  $m$  goes down, then the acceleration  $a$  goes up. As  $m$  goes to zero,  $a$  goes to infinity, unless  $F = 0$ . Basically, as soon as you try to have a non-zero angle, the tip of the string responds so quickly that the end stays flat.

This is the key difference between a fixed string and an open (freely sliding) string. For a fixed string, the height at the end stays pegged at zero. For a freely sliding string, the *slope* at the end stays pegged at zero. Whatever our string is going to do still has to be a solution to the wave equation, but now we need to arrange our signals to cancel out the slope at the end. We saw to keep the height equal to zero we sent a copy of our incoming signal back, but flipped around and upside down. To keep the slope equal to zero, we flip our signal around left-right, but do *not* flip it upside down.

To understand why this works, picture yourself climbing up a mountain. The peak is in front of you, the parking lot is behind you, and there's some slope where you are. Now if you flip the mountain around but you stay in the same spot, you're now climbing down the mountain. The peak is behind you, the parking lot is in front of you, and so whatever your slope was before, it's now negative. If I add my original mountain plus my flipped mountain, the slopes have to cancel and so I'm left with zero slope. You can see this in action [on desmos here](#). The incoming wave is red, our cancelling wave is green, and the sum is purple. If you look at  $x = 0$ , you see that the sum of the waves is always flat. With a fixed end, we get a *negative* reflection at the end. With a free end, we get a *positive* reflection.

### 2. Standing Waves with Free End

For music, what we want to know is at what frequencies will our string vibrate? If you look at [this plot](#), we can figure this out. First, picture a standing wave on a string that goes from 0 to  $\pi$ .

We can have any number of half-wavelengths on our string, so we get all integer multiples of our fundamental as harmonics. We've seen this plot before, but now I've added an extra dot in the middle of the string. If you start with  $k = 1$ , you can see that the string stays at zero at the ends as expected, and reaches a largest/most negative value in the middle. You'll see that the string is always flat at the half-way point<sup>1</sup>. That means this wave works for a string of length  $\pi$  that is fixed at both ends, *and* for a string of length  $\pi/2$  that is fixed at one end, and open at the other. The wavelength remains the same ( $2\pi$ ), but our string got half as long. If our string is  $\pi/2$  long, and the wavelength is  $2\pi$ , then the length of the string is a *quarter* of the fundamental. Alternatively, if I have two strings of the same length  $L$ , the fundamental of the string fixed at both ends corresponds to a wave with length  $2L$ , while the open string corresponds to a wave with length  $4L$ . The open string will be exactly an octave lower than the fixed string.

So, a string open at one end behaves just like a string fixed at both ends that's twice as long? Not so fast! Next, switch to  $k = 2$  in the desmos plot. The sine wave stays zero at 0 and  $\pi$ , as expected. However, what happens in the middle? Now the string goes to zero there as well, and the slope is large. This is *not* allowed because the slope has to go to zero for our half-length string. That means we do not get the  $k = 2$  harmonic. If we go to  $k = 3$ , though, we're back to having an odd number of half-wavelengths and so the string is flat in the middle. If you play with the plot/think about it, I hope you can convince yourself that whenever  $k$  is *even*, the slope does not go to zero in the center, and so even  $k$ 's are not allowed for a string open at one end. Whenever  $k$  is *odd*, then the string is indeed flat, and so those values of  $k$  are allowed.

### 3. Open at Both Ends

For completeness, let's consider what happens when a string is free at both ends. If we have a string going from 0 to  $\pi$ , then  $\cos(x)$  will be flat at 0 (where  $\cos = 1$ ), and flat at  $\pi$  (where  $\cos = -1$ ). What happens if I set  $k = 2$  for  $\cos(2x)$ ? Well,  $\cos(0)$  is still 1, so we remain flat at  $x = 0$ . At  $x = \pi$ ,  $\cos(2\pi) = 1$ , so again we're flat. The  $k = 1$  case corresponds to half a wavelength on the string, the  $k = 2$  case is a full wavelength, and so on. Any integer value of  $k$  will make  $\cos(kx)$  flat at  $x = 0$  and  $x = \pi$ , so we can put any number of half-wavelengths on the string/have any integer value of  $k$ . That means that a string with *two* free ends act exactly like a string with *no* free ends. We get all the harmonics, and the fundamentals are the same frequency.

### 4. Summary and Discussion

Here's a quick recap of the behavior of strings with fixed or free ends. A string with fixed ends can have any number of half-wavelengths on the string. The fundamental wavelength is *twice* the string length, and the frequency of the fundamental is  $\nu_f = v/2L$  where  $v$  is the speed of waves on the string, and  $L$  is the length of the string. A string that has one end fixed and one end free has

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<sup>1</sup>If you have a slope, you can move uphill by going in the uphill direction. If you're at the top of a hill, the slope has to be zero because otherwise there would still be a direction you could move in to go up.

fundamental  $\nu_f = v/4L$  and only odd harmonics; we get  $\nu = (1, 3, 5, \dots)\nu_f$ . A string free at both ends behaves just like a string with fixed ends.

While we've worked the behaviour out for strings, the behavior of straight pipes like flutes is exactly the same. A flute is closed at one end and open at the other, so it will behave exactly like a string with one free end. We'll see *why* this is later on<sup>2</sup>. The wave equation is very simple, so I hope it's at least not surprising that a flute that is closed at one end and open at the other will behave like a string that is fixed at one end and free at the other. See below for plots showing the strength of harmonics for a flute. The even harmonics on the flute aren't exactly zero, but they are weaker than the odd ones.

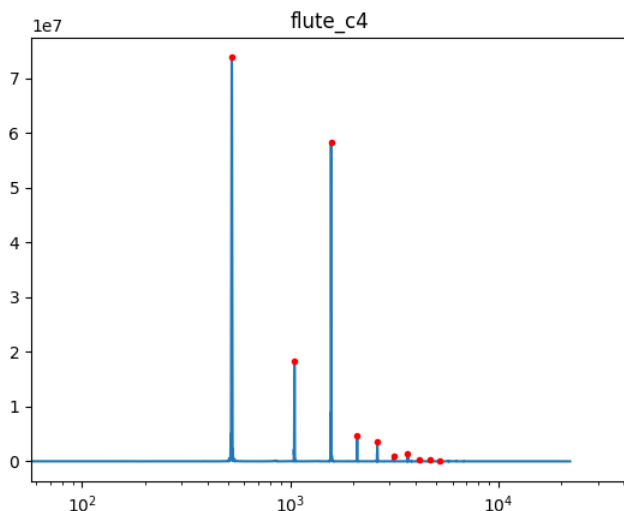


Fig. 1.— Sound from a flute playing a concert C4. The plot shows loudness of the flute vs. frequency. The largest spike at the lowest frequency is the fundamental. The red dots are frequencies that are integer multiples of the fundamental. The second  $k = 2$  harmonic is much lower than the first or the third. It's not exactly zero, because instruments are complicated and rarely exactly the perfectly ideal simple physical system, but the second harmonic is much lower than the first or third.

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<sup>2</sup>At the closed end, the air can't move because the end of the flute is in the way. At the open end, I can't have any force on the air because otherwise it would move in response to the force. We'll see that statement is equivalent to having a free end on a string.